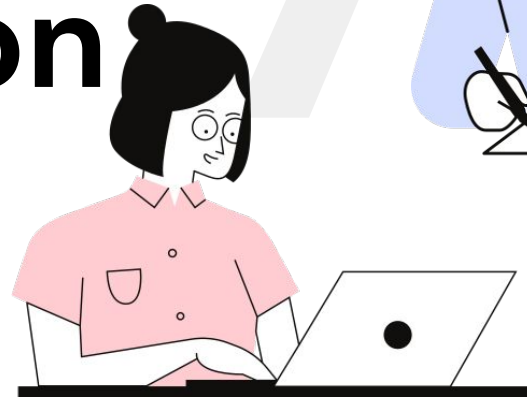




BRITISH INTERNATIONAL
SCHOOL OF SULAYMANIYAH

January
2025

Information & Communication Technology



OBJECTIVES

- ❖ **Understand** the analysis stage, including system analysis methods, identifying inputs/outputs, and determining hardware/software needs.
- ❖ **Learn** the design stage elements, such as file structures, input/output formats, data capture forms, screen/report layouts, and validation routines.
- ❖ **Explore** the development and testing stage, covering test strategies, test plans, and different types of test data.
- ❖ **Examine** the implementation stage, including direct changeover, parallel running, pilot running, and phased implementation.
- ❖ **Understand** the documentation and evaluation stages, including technical/user documentation and comparing the new solution to the original system.

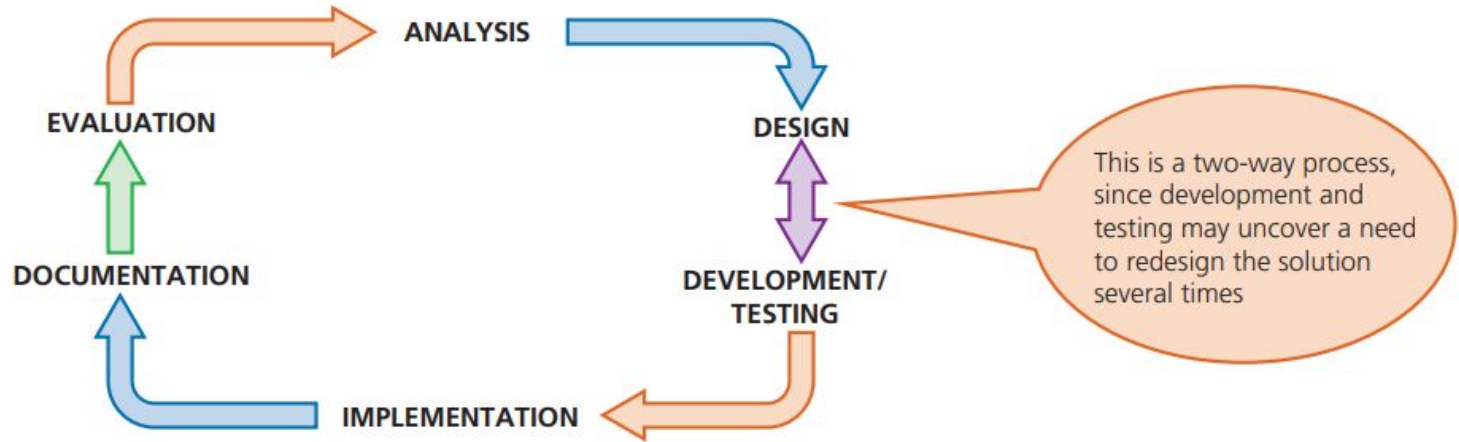
System Upgrade: Reasons & Process

A company may need to upgrade its computer system due to:

- ❖ **Obsolescence** – Existing equipment cannot be repaired.
- ❖ **Legal/tax changes** – New laws require system updates.
- ❖ **Better hardware available** – Enhances efficiency & reliability.
- ❖ **Business expansion** – Requires improved system capabilities.

- ❖ Oversees the upgrade process.
- ❖ Analyzes the current system.
- ❖ Suggests improvements, calculates costs, and reports advantages.
- ❖ Ensures smooth installation, evaluation, and modifications.

- ❖ **Analysis** – Understanding current system requirements.
- ❖ **Design** – Creating a new system plan.
- ❖ **Development & Testing** – Implementing & verifying system functions.
- ❖ **Implementation** – Installing and integrating the new system.
- ❖ **Evaluation & Maintenance** – Reviewing and making necessary updates.



- ❖ **Research current system** – Identify processes, inputs, outputs.
- ❖ **Identify problems** – Understand system weaknesses.
- ❖ **Determine user & information requirements.**
- ❖ **Recommend necessary hardware & software.**

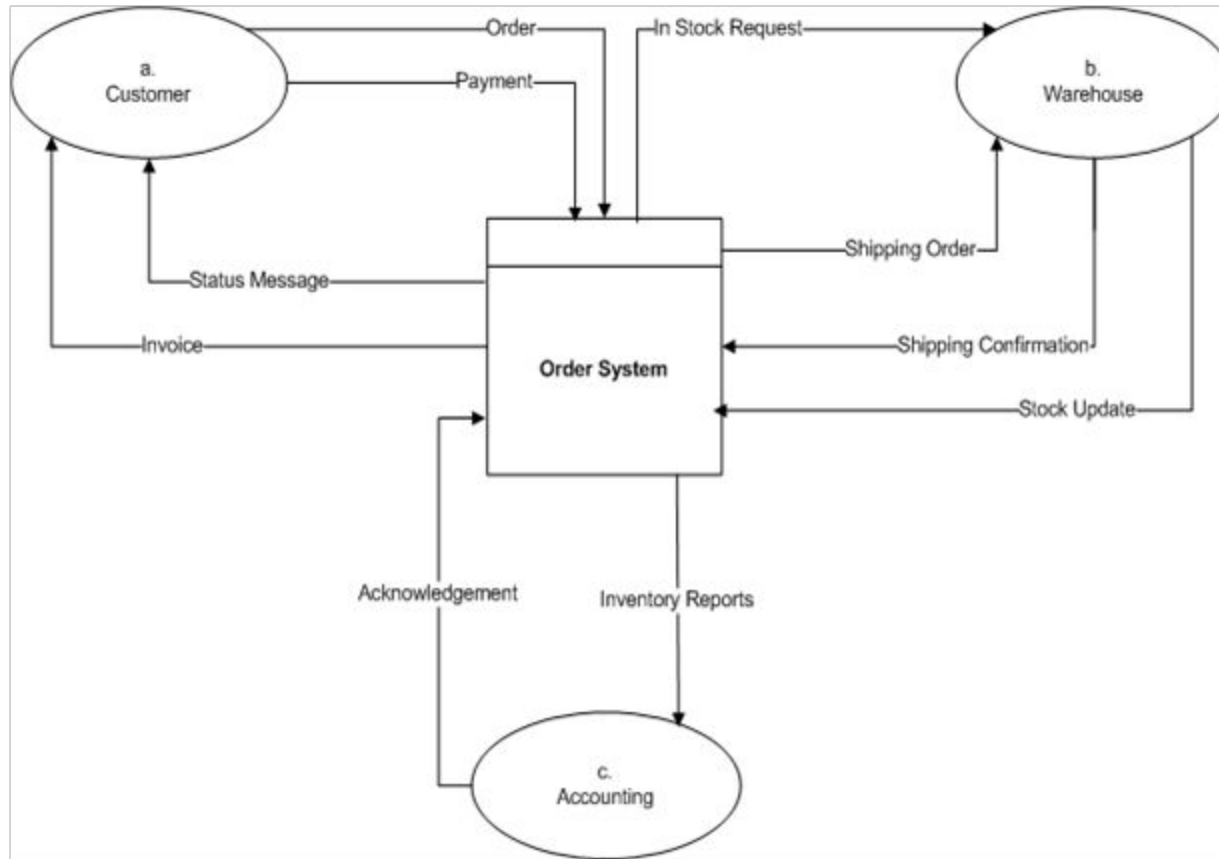
- ❖ Observation – Watching users interact with the system.
 - Reliable data, overall system view, cost-effective.
 - Users may alter behavior when observed (Hawthorne effect).
- ❖ Interviews – One-on-one discussions with users.
 - In-depth responses, flexible questioning.
 - Time-consuming, costly, possible bias in answers.
- ❖ Questionnaires – Distributed surveys to users.
 - Quick, cost-effective, allows anonymity.
 - Low response rate, rigid questions, no follow-ups.
- ❖ Existing Document Examination – Reviewing files, manuals, and reports.
 - Provides unique insights.
 - Time-consuming and expensive.

Identify:

- ❖ **Inputs** – Data entry methods.
- ❖ **Outputs** – System-generated results.
- ❖ **Processing** – System operations and transformations.
- ❖ **Problems** – Existing system inefficiencies.
- ❖ User & information requirements.

Flight Booking System (Data Flow Analysis)

- ❖ **Input:** Passenger details (name, passport, flight date).
- ❖ **Processing:** Checking flights, verifying data, assigning booking reference.
- ❖ **Output:** E-ticket, confirmation message.
- ❖ **Storage:** Database to keep booking records.



- ❖ Identifying Problems & Requirements
 - Analysts examine Data Flow Diagrams (DFD) to identify bottlenecks, errors in the current system, and determine user & information requirements.
- ❖ User Requirements
 - Bridge gap between business managers and system developers.
 - Written in natural language to ensure clarity.
 - Ensures the proposed system meets business needs, including accessibility.
- ❖ Information Requirements
 - What? (Data needed) & When? (Timing of data processing)
 - Used to create a functional requirements specification:
 - Defines customer, vendor, and employee interactions with the system.

Hardware Justification:

❖ Input Devices:

- Barcode readers – Efficient, reduces errors.
- Scanners – Convert paper documents to digital format.
- Touch screens – User-friendly for customer interactions.

❖ Output Devices:

- 3D printers – Cost-effective for prototyping.
- Large monitors – Needed for CAD design.
- Speakers – Assist employees with disabilities.

- ❖ Operating System – Chosen based on company needs.
- ❖ Application Software:
 - Off-the-shelf software – Cost-effective, minimal training needed.
 - Bespoke software – Tailored to needs but costly and time-intensive.
- ❖ Storage Considerations:
 - Size – Future-proofing required.
 - HDDs or SSDs – Based on access speed and durability.
 - Magnetic Tape – Suitable for large archives with minimal access needs.

S25_QP11_Q10

During the analysis of a current system, research needs to be carried out.

(a) **Describe** the characteristics of each of the following research methods

(i) Observation

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(ii) Interviews

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.....[2]

Question	Answer	Marks
10(a)(i)	Two from: Involves watching people use the current system to see how it works / errors in the process Allows the systems analyst to see the input, process and output of the current system Due to the Hawthorne effect the results may not be reliable	2
10(a)(ii)	Two from: One to one method of question and answer // Face to face conversation Its between the analyst and employee Allows the analyst to find more information about a particular topic Allows the interviewer to evaluate the answer	2

Design Process Includes:

- ❖ **File & Data Structures** – Organizing data for efficiency.
- ❖ **Input Formats** – Designing user-friendly data entry methods.
- ❖ **Output Formats** – Structuring screen and report layouts.
- ❖ **Validation Routines** – Ensuring data integrity.

- ❖ Data is stored in records, each containing multiple fields.
- ❖ Primary Key – Uniquely identifies each record.

Example File Structure:

- ❖ **Fields:** Product Code, Year of Manufacture, Description, Price, Department.
- ❖ **Field Types:** Text, Numeric, Currency, Character.

- ❖ **Alphanumeric** – Mix of letters/numbers (e.g., A345FF).
- ❖ **Character** – Single letter/text (e.g., X).
- ❖ **Text** – String of characters (e.g., "example_text").
- ❖ **Boolean** – Logical values (Yes/No).
- ❖ **Numeric** – Used for calculations (Integer, Decimal, Currency).
- ❖ **Date/Time** – Stores date and time values.

Optimizing Field Lengths:

- Prevents wasted memory.
- Example: If max data is 5 characters, avoid setting field size to 50.

- ❖ Validation ensures data input meets predefined criteria but does not check accuracy.

Common Validation Checks:

- **Range Check** – Ensures input falls within a set range ($120 < \text{age} > 0$).
- **Type/Character Check** – Verifies data type (names w/o numbers).
- **Length Check** – Ensures input has a fixed number of characters (8-character password).
- **Format Check** – Ensures data follows a specific structure (date dd/mm/yyyy).
- **Presence Check** – Ensures required fields are not left empty (phone number).
- **Check Digit** – Extra digit added to detect errors in numeric sequences (preventing typos in ID numbers).

❖ Paper-Based Forms:

- Clear headings, text boxes, and tick boxes for easy data entry.
- Use of character boxes for structured input (e.g., names, phone numbers).
- Sufficient space for answers and readable fonts/colors.

❖ Computer-Based Forms:

- Text boxes, drop-down menus, and radio buttons for structured input.
- On-screen help & validation to guide users.
- Double-entry verification for key data like email.
- Control buttons (e.g., "Save", "Next", "Clear").
- Example (Car Sales Form Changes):
- Make of car → Drop-down menu for manufacturers.
- Date registered → Drop-down boxes for day, month, year.
- Price → Text box with validation.
- New/Used → Tickbox or radio button.

Screen Layouts:

- ❖ Correct field sizes for clarity.
- ❖ Clear instructions & labels for users.
- ❖ Full utilization of screen space to avoid blank areas.
- ❖ Appropriate fonts, colors, and contrast for readability.

Report Layouts:

- ❖ Includes headers, footers, and proper formatting.
- ❖ Data presented in tables for clarity (e.g., list of sales managers over 40).
- ❖ Ensures search criteria results are accurately displayed.

W25_QP11_Q3

A headteacher has decided that all pupil reports are to be typed into a computer. He has asked the computer technician to design a layout for a pupil report.

- (a) **Describe** five features of a well-designed report layout.

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Question	Answer	Marks
3(a)	Five from: Headings / subheadings relate to the content Headings / subheadings are clear and easy to read Font style / type is easy to read Font size is large enough to be easily read Makes good use of the page area / white space Use of headers and footers to help in the layout Contrasting colours to make it easy to read All data is fully visible	5

Need for Testing

- ❖ Ensures system functions correctly before going live.
- ❖ File Structure Finalization: determines data types, field lengths, key fields, and data linking.
- ❖ Validation & Verification: ensures accurate data transfer from paper-based to electronic systems.
- ❖ User Interface Testing: evaluates input/output devices and accessibility features (e.g., assistive technology).

- ❖ **Data Structures** – Ensures data is stored correctly.
- ❖ **File Structures** – Checks file functionality and data retrieval.
- ❖ **Input Formats** – Tests correct data entry.
- ❖ **Output Formats** – Verifies clarity and accuracy of outputs.
- ❖ **Validation Routines** – Ensures invalid data is rejected.

- ❖ Modular Testing:
 - Software is broken into smaller modules and tested separately.
 - Each module is modified if errors occur and re-tested.
- ❖ System Testing:
 - Once all modules are combined, the entire system is tested to detect compatibility issues.

- ❖ A test plan includes:
 - List of tests, expected outcomes, and comparison with actual results.
 - Types of test data:
 - **Normal Data** – Expected inputs within valid range (e.g., month: 1-12).
 - **Extreme Data** – Boundary values (e.g., month: 1 or 12).
 - **Abnormal Data** – Invalid inputs causing errors (e.g., letters in a numeric field).

- ❖ Live Data Testing:
 - Uses real data to compare new vs. existing system results.
 - If discrepancies arise, system modifications are required.

- ❖ Valid Range: 21°C to 50°C
- ❖ Expected Behavior:
 - **Accepted:** Normal (35, 45), Extreme (21, 50).
 - **Rejected:** Abnormal (e.g., "fifty", 20, -5, 100.5).

If actual results differ, system modifications are needed.

S25_QP13_Q4

A software developer is producing a new app. The app will need to be tested before it is implemented

(a) **State** why the following elements need to be tested.

(i) Data structure

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.....[1]

(ii) Input format

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.....[1]

Question	Answer	Marks
4(a)(i)	One from: To ensure they store data correctly To identify any errors It's crucial for creating robust apps	1
4(a)(ii)	One from: To ensure the user enters data correctly	1

Implementation Stages

- ❖ **Data Transfer:** Key in data, scan documents, transfer paper/electronic files.
- ❖ **Staff Training:** Essential to ensure smooth transition

Changeover Methods

- ❖ Direct Changeover.
- ❖ Parallel Running
- ❖ Pilot Implementation
- ❖ Phased Implementation

Direct Changeover

- ❖ Old system stops, new system starts immediately.
- ❖ Immediate benefits, lower cost ✓
- ❖ Risky—failure can be disastrous ✗

Parallel Running

- ❖ Old and new systems run together for a period.
- ❖ Safer—old system acts as backup ✓
- ❖ Costly and time-consuming (data entered twice) ✗

Pilot Implementation

- ❖ New system tested in one department before full rollout.
- ❖ Limits failure impact, cost-effective training ✓
- ❖ Affected department may face challenges ✗

Phased Implementation

- ❖ System introduced in parts over time.
- ❖ Allows gradual adaptation, minimizes risk ✓
- ❖ Slow process, more expensive than direct changeover ✗

M25_QP12_Q7

A company is planning to implement a new computer software system. It is going to use direct changeover. One of the reasons for using this method is that the benefits are immediate.

- (a) **Discuss** other benefits and drawbacks of using this method of implementation.

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Question	Answer	Marks
7(a)	Benefits Max three from: Cheaper staffing costs Only needs to be one set of staff Only one system in place therefore cheaper The software is fully tested before implementation therefore fewer errors Prevents the duplication of data Saves time on entering data as data is entered once Drawbacks Max three from: Staff have very little time to train on the new system as the system is implemented immediately The original system cannot be used If the new system fails, all data is lost Increase staff costs as staff will need to be trained to use the new system Staff cannot train easily on the live system	4

Technical Documentation (for programmers/analysts)

- ❖ **System details:** Code, programming language, flowcharts, algorithms.
- ❖ **Hardware & software requirements:** Minimum specs needed.
- ❖ **File structures:** Database format, validation rules.
- ❖ **Known bugs & error messages:** Troubleshooting support.

User Documentation (for end-users)

- ❖ **System operation:** Installation, saving, searching, sorting, printing.
- ❖ **User interface:** Input formats, screen layouts, print layouts.
- ❖ **Troubleshooting:** Error messages, FAQs, help guides.
- ❖ **Training materials:** Tutorials, login/logout procedures.

S25_QP13_Q5

Following the development and testing of a new computer program, user documentation is produced.

- (a) **Explain** why user documentation is needed.

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.....[1]

- (b) The input format is one component in user documentation.

State three other components found in user documentation.

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.....[3]

Question	Answer	Marks
5(a)	One from: To help the user learn how to use the program To help with troubleshooting	1
5(b)	Three from: Error handling Error messages Frequently asked questions / FAQ Glossary of terms Hardware requirements How to add records How to delete / edit records How to load / run / install the program How to print data How to save a file Limitations of the system Output format Purpose of the system Sample runs Software requirements Troubleshooting guide / helpline	3

Assessing System Performance

- ❖ Compare final system with initial requirements.
- ❖ Identify limitations & possible improvements.
- ❖ Gather user feedback (interviews, questionnaires).
- ❖ Compare new vs. old system performance:
 - Task completion time
 - Ease of use
 - Business efficiency improvements

Potential Updates

- ❖ **Hardware updates:** New technology, business needs, user feedback.
- ❖ **Software updates:** User feedback, company changes, legal requirements.

S25_QP13_Q9

A programmer has created a database for Tawara International Stores. One of the fields is Country_name which stores data as a three-letter code. For example, NZL is the code for New Zealand.

- (b) The database needs to be evaluated. Part of the evaluation is to identify any limitations.

Describe four other tasks carried out during the evaluation.

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9(b)	Four from, for example: Compare test results from the new system with the old Comparison of the final solution with the original user requirements Identifies improvements to the system Evaluate the user's responses to using the system Measure the time taken to complete the task and compare with the old system The appropriateness of the solution The efficiency of the solution	4
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